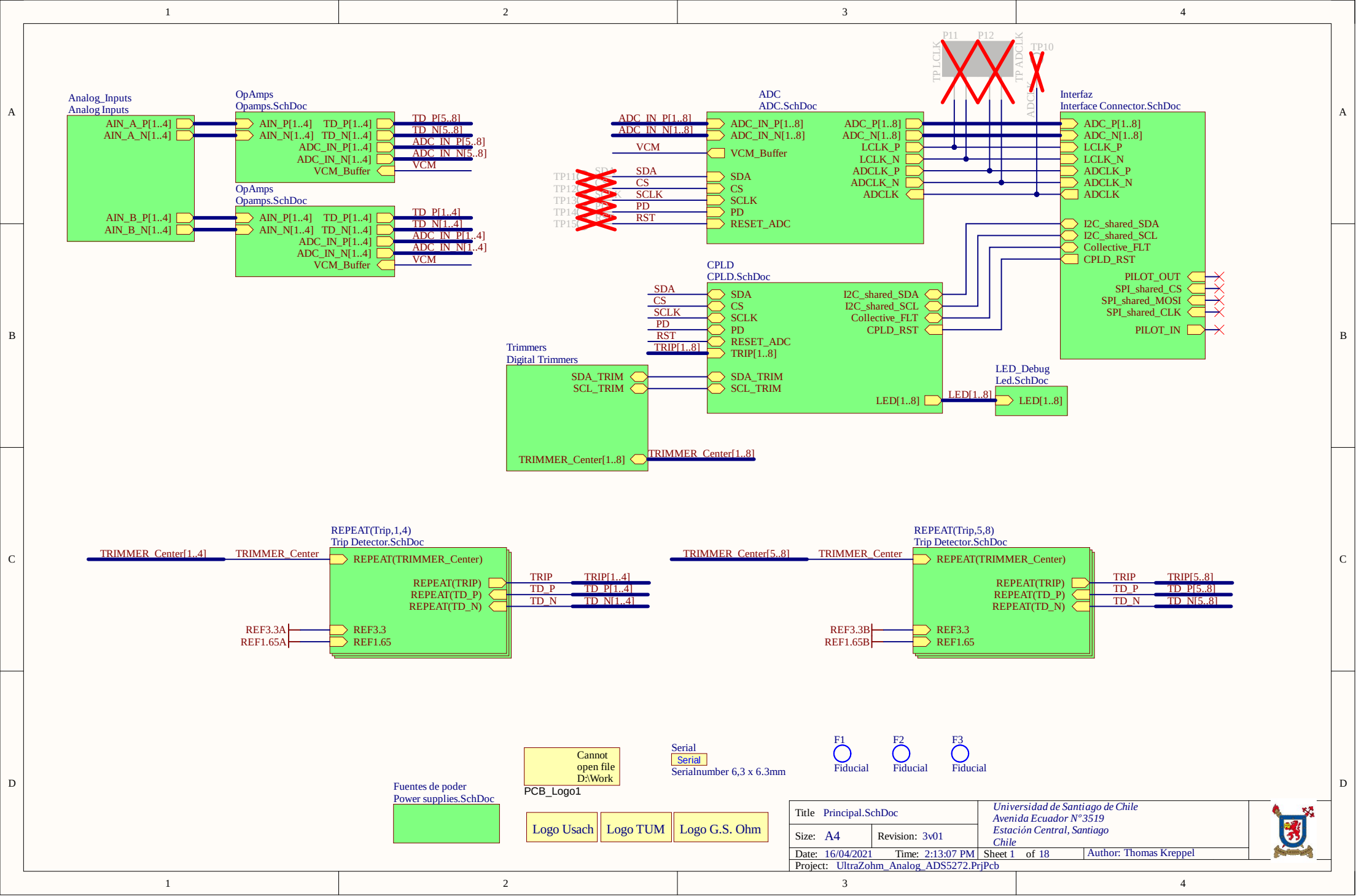
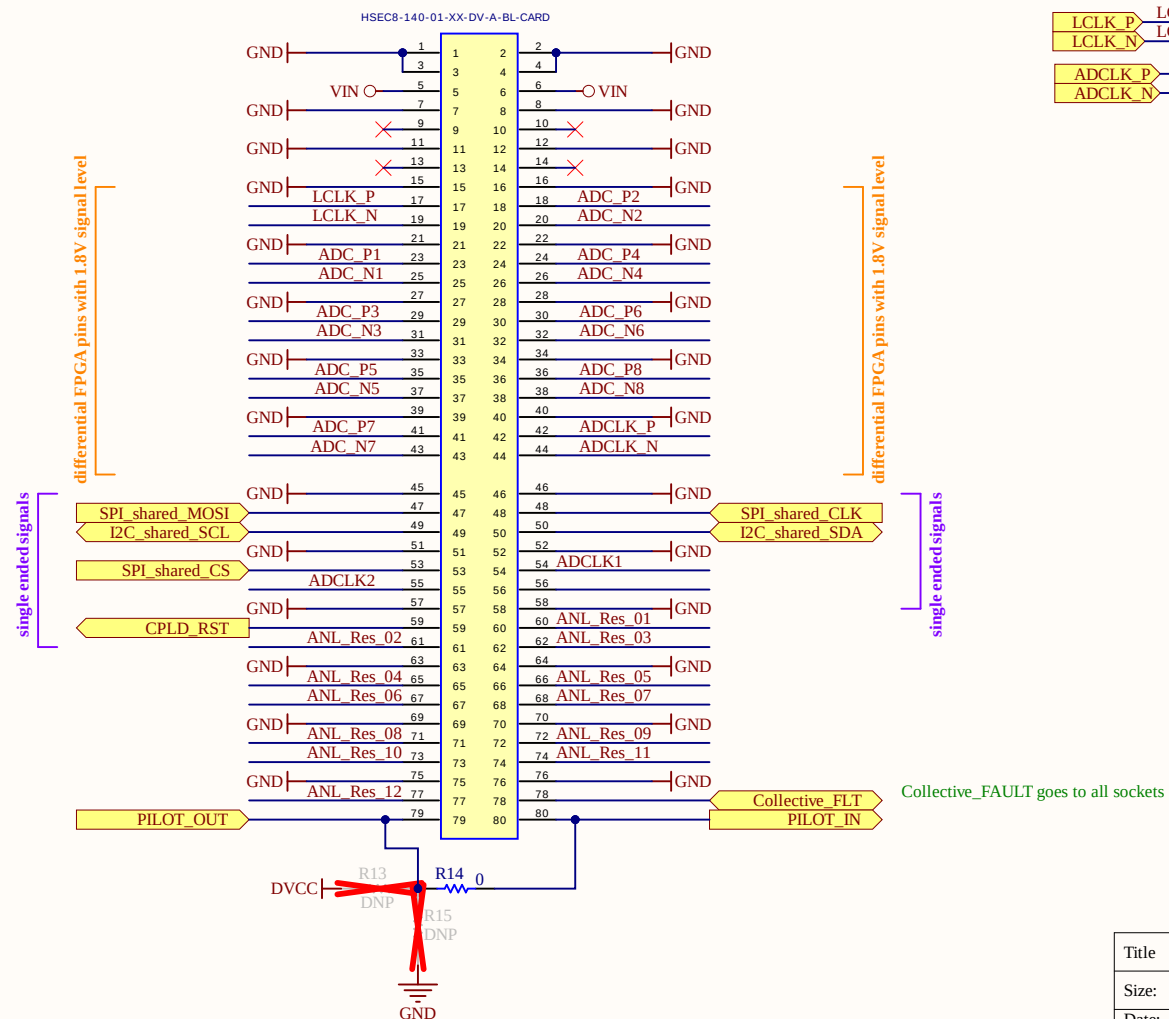




A

A



C

C

D

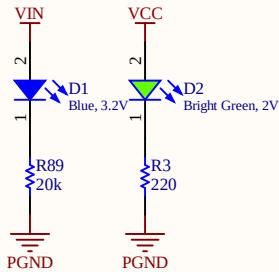
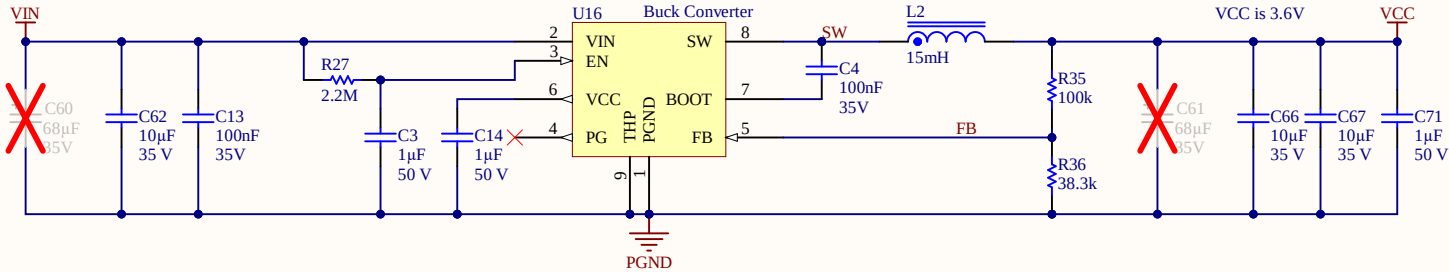
D

Title: Interface Connector.SchDoc		<i>Universidad de Santiago de Chile</i> <i>Avenida Ecuador N° 3519</i> <i>Estación Central, Santiago Chile</i>		
Size: A4	Revision: 3v01			
Date: 16/04/2021	Time: 2:13:07 PM	Sheet 2 of 18	Author: Thomas Kreppel	
Project: UltraZohm_Analog_ADSS272.PriPcb				

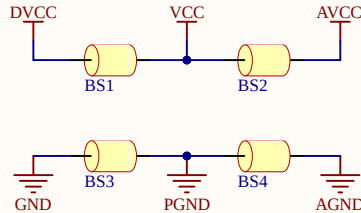
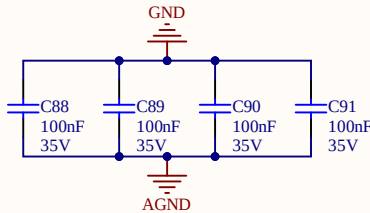
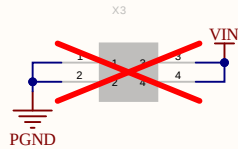
VIN between 12V and 24V

R35 = 100k, Vout = 3.6V, Vref = 1V

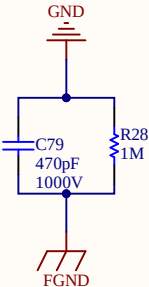
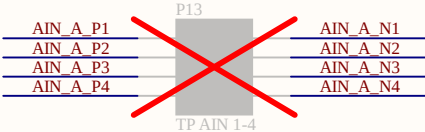
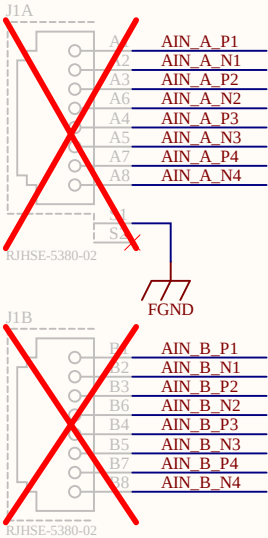
$$R36 = \frac{R35}{\frac{Vout}{Vref} - 1} = \frac{100k}{\frac{3.6V}{1V} - 1} = 38.3k$$



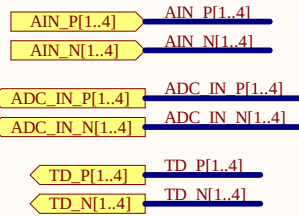
LED has forward voltage of 3.2V, thus the resistor is calculated as (V-3.2)/I with V being the supply voltage and I the forward current. A forward current of 1mA was assumed for this calculation



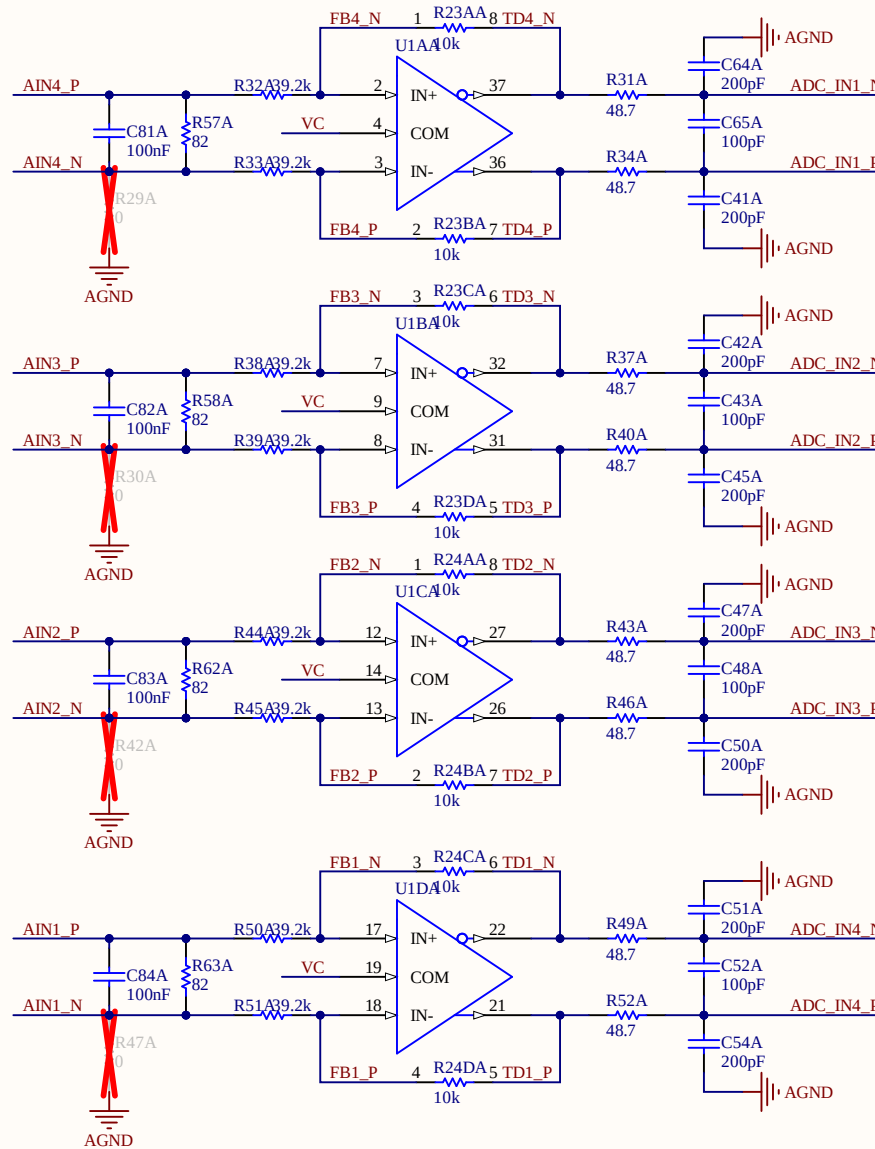
Input voltage between +-4V



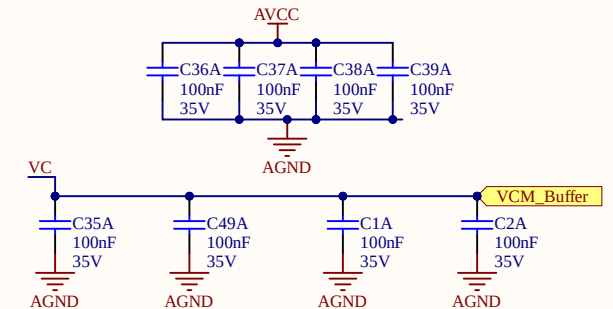
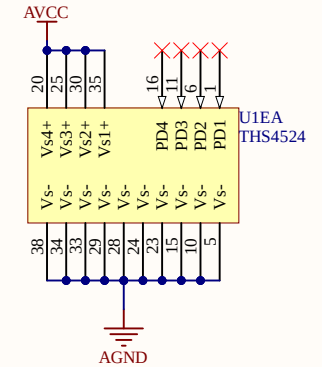
G=0.25



AIN_P4	AIN4_P
AIN_N4	AIN4_N
AIN_P3	AIN3_P
AIN_N3	AIN3_N
AIN_P2	AIN2_P
AIN_N2	AIN2_N
AIN_P1	AIN1_P
AIN_N1	AIN1_N
ADC_IN_P4	ADC_IN4_P
ADC_IN_N4	ADC_IN4_N
ADC_IN_P3	ADC_IN3_P
ADC_IN_N3	ADC_IN3_N
ADC_IN_P2	ADC_IN2_P
ADC_IN_N2	ADC_IN2_N
ADC_IN_P1	ADC_IN1_P
ADC_IN_N1	ADC_IN1_N
TD_P4	TD4_P
TD_N4	TD4_N
TD_P3	TD3_P
TD_N3	TD3_N
TD_P2	TD2_P
TD_N2	TD2_N
TD_P1	TD1_P
TD_N1	TD1_N
FB4_P	
FB4_N	
FB3_P	
FB3_N	
FB2_P	
FB2_N	
FB1_P	
FB1_N	



Filtro Antialiasing:
 Considerando una frecuencia de corte de 16.25MHz (1/4 de 65MSample/seg)
 Se debe utilizar una R de 98ohm y un C de 100pF
 Como el filtro es diferencial, la resistencia se calcula a la mitad, es decir 49ohm



R29,R30,R42,R57 can be fitted with 0R if signal is single ended, to pull negative input to ground

Title Opamps.SchDoc

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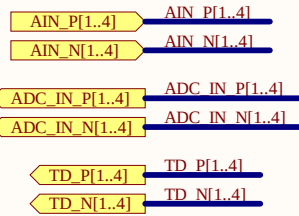
Author: Thomas Kreppel

Project: UltraZohm_Analog_ADS5272.PrjPcb

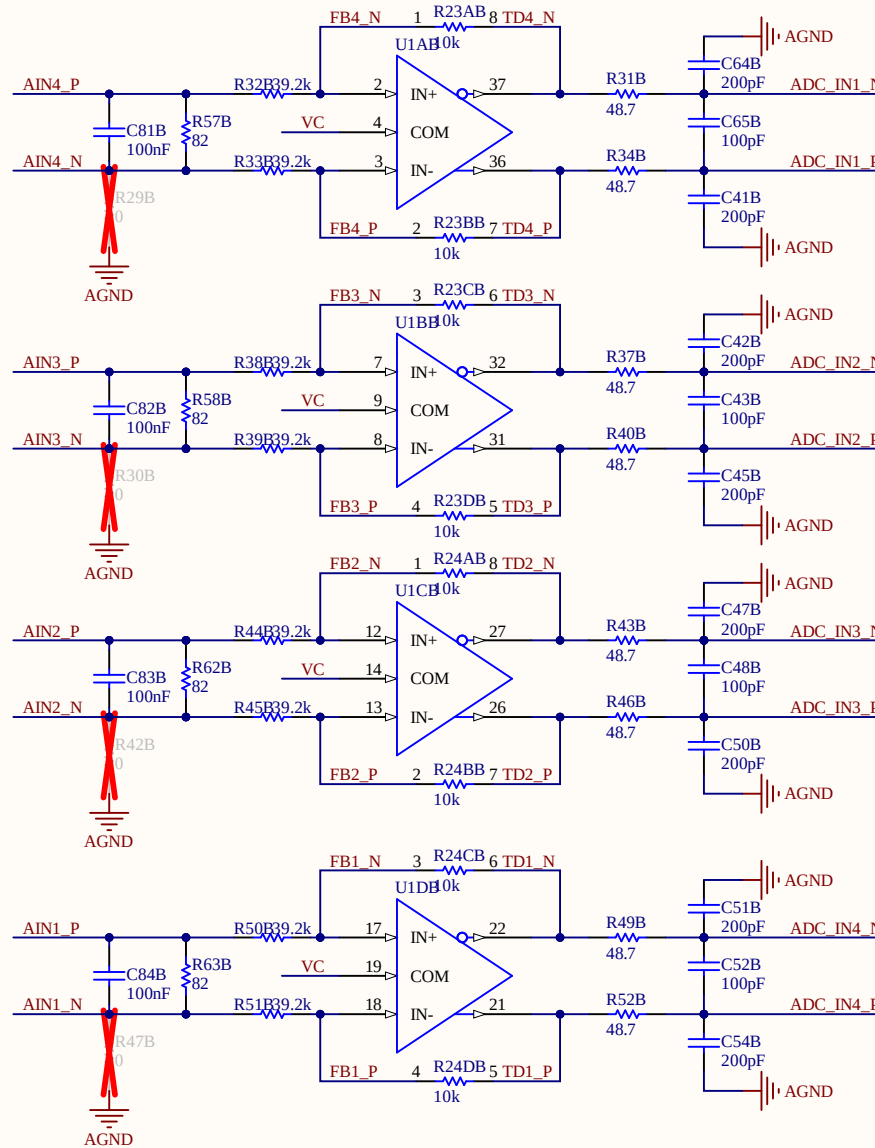
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 Estación Central, Santiago
 Chile



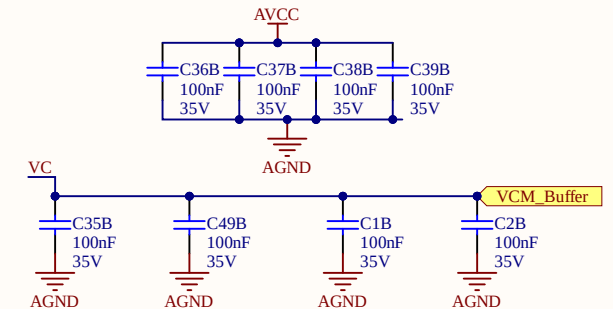
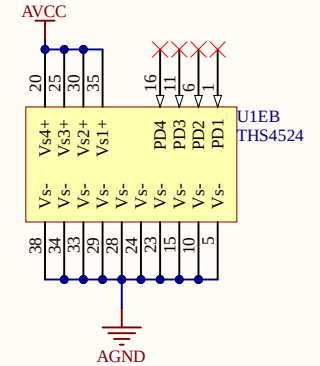
G=0.25



AIN_P4	AIN4_P
AIN_N4	AIN4_N
AIN_P3	AIN3_P
AIN_N3	AIN3_N
AIN_P2	AIN2_P
AIN_N2	AIN2_N
AIN_P1	AIN1_P
AIN_N1	AIN1_N
ADC_IN_P4	ADC_IN4_P
ADC_IN_N4	ADC_IN4_N
ADC_IN_P3	ADC_IN3_P
ADC_IN_N3	ADC_IN3_N
ADC_IN_P2	ADC_IN2_P
ADC_IN_N2	ADC_IN2_N
ADC_IN_P1	ADC_IN1_P
ADC_IN_N1	ADC_IN1_N
TD_P4	TD4_P
TD_N4	TD4_N
TD_P3	TD3_P
TD_N3	TD3_N
TD_P2	TD2_P
TD_N2	TD2_N
TD_P1	TD1_P
TD_N1	TD1_N
FB4_P	
FB4_N	
FB3_P	
FB3_N	
FB2_P	
FB2_N	
FB1_P	
FB1_N	



Filtro Antialiasing:
 Considerando una frecuencia de corte de 16.25MHz (1/4 de 65MSample/seg)
 Se debe utilizar una R de 98ohm y un C de 100pF
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R29,R30,R42,R57 can be fitted with 0R if signal is single ended, to pull negative input to ground

Title Opamps.SchDoc

Size: A4

Revision: 3v01

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Time: 2:13:09 PM

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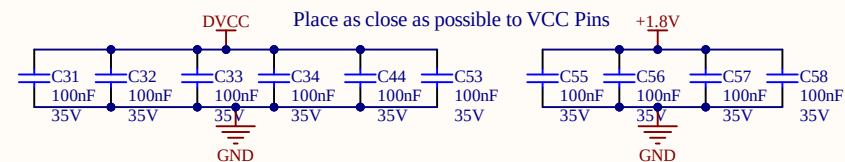
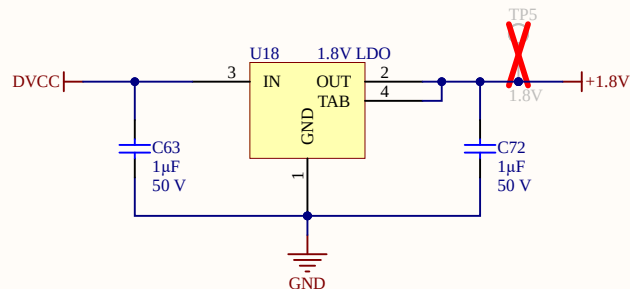
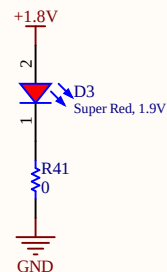
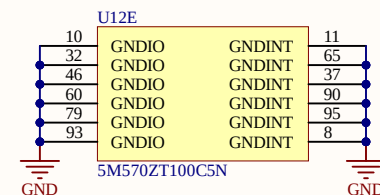
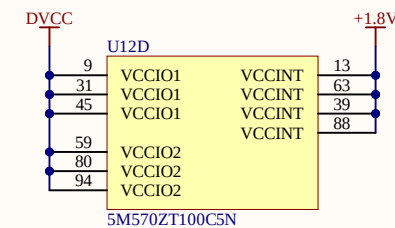
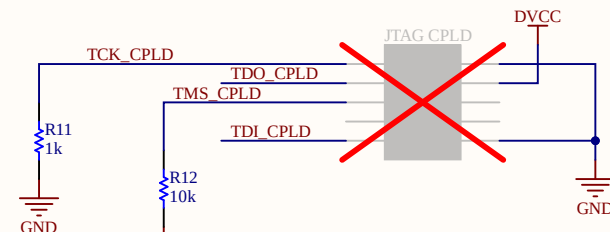
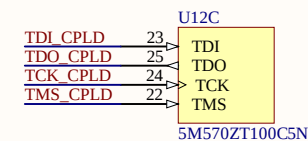
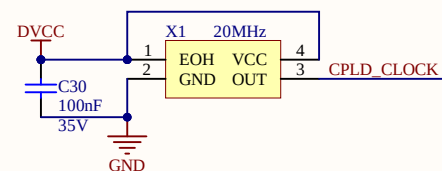
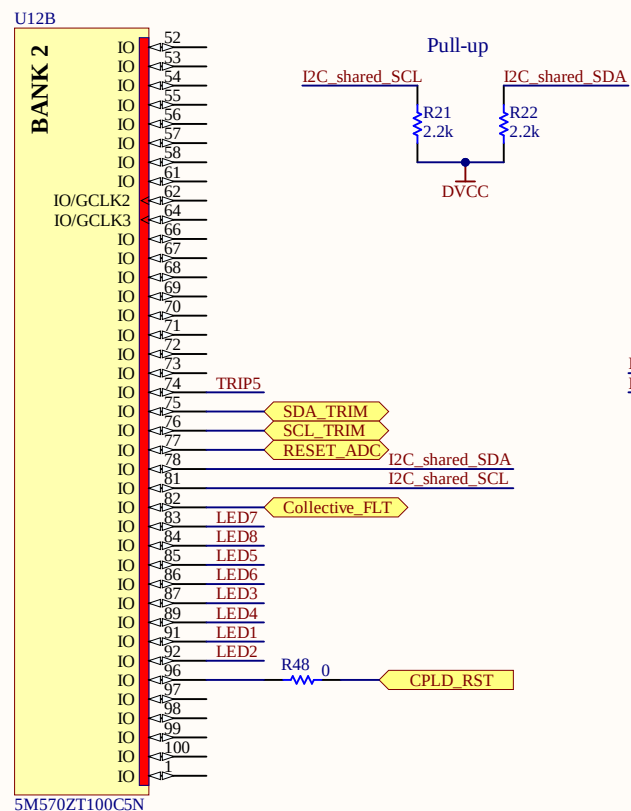
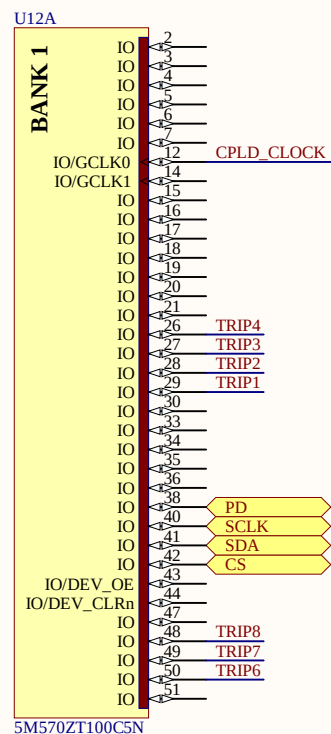
Author: Thomas Kreppel

Project: UltraZohm_Analog_ADS5272.PrjPcb

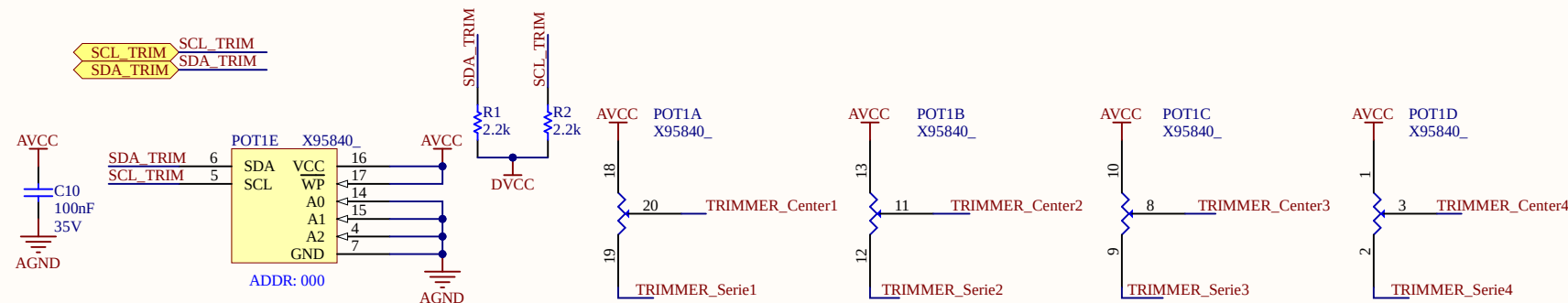
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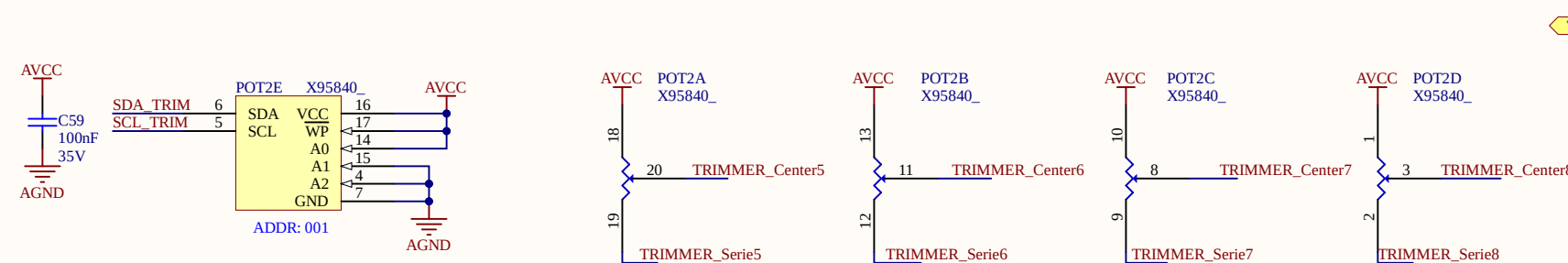




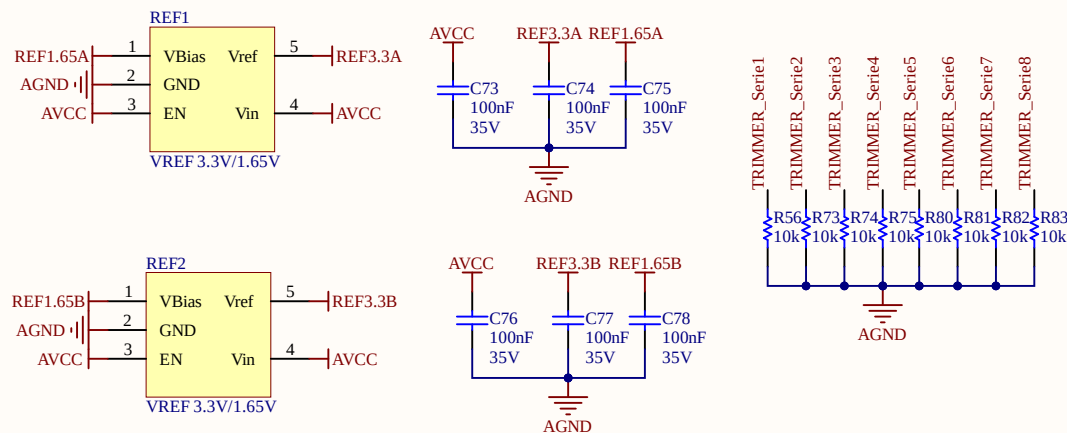
A



B



C



D



A

A

B

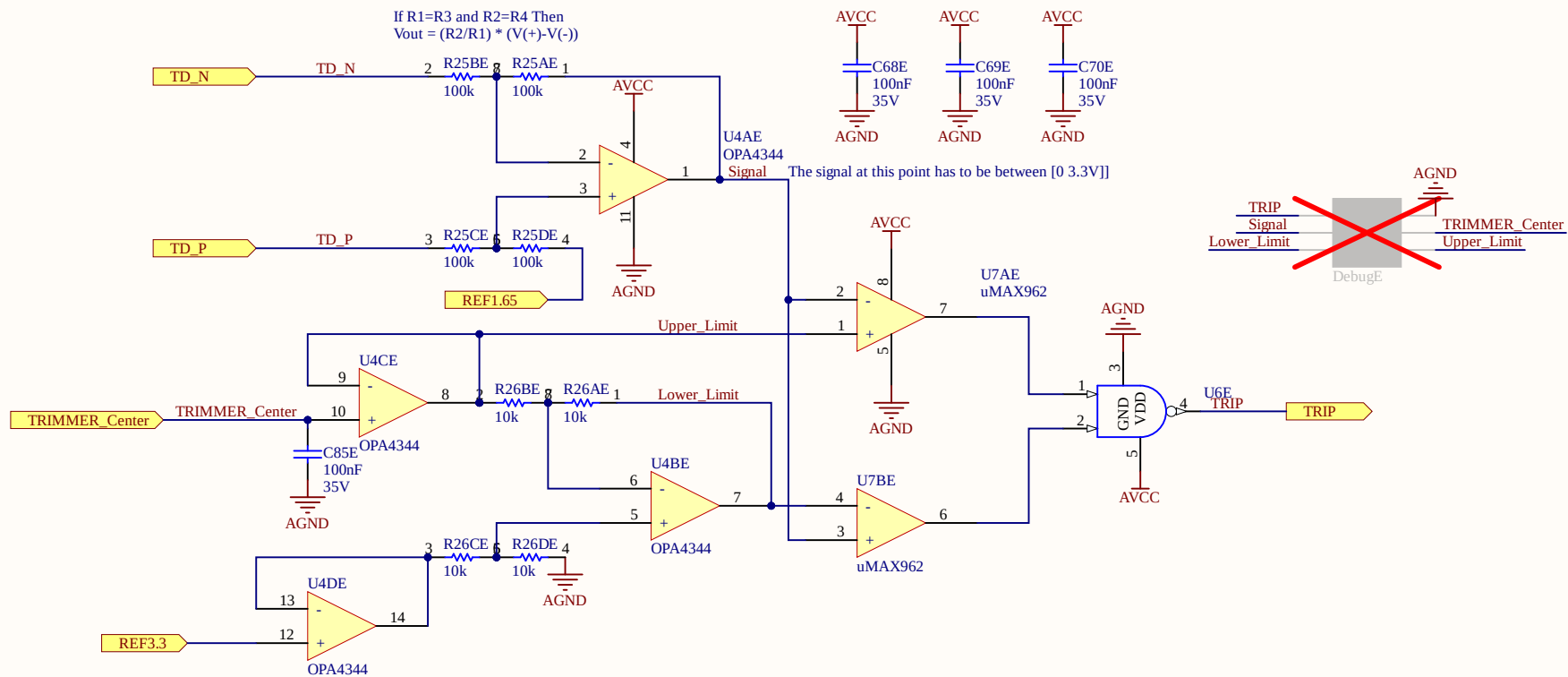
B

C

C

D

D



A

A

B

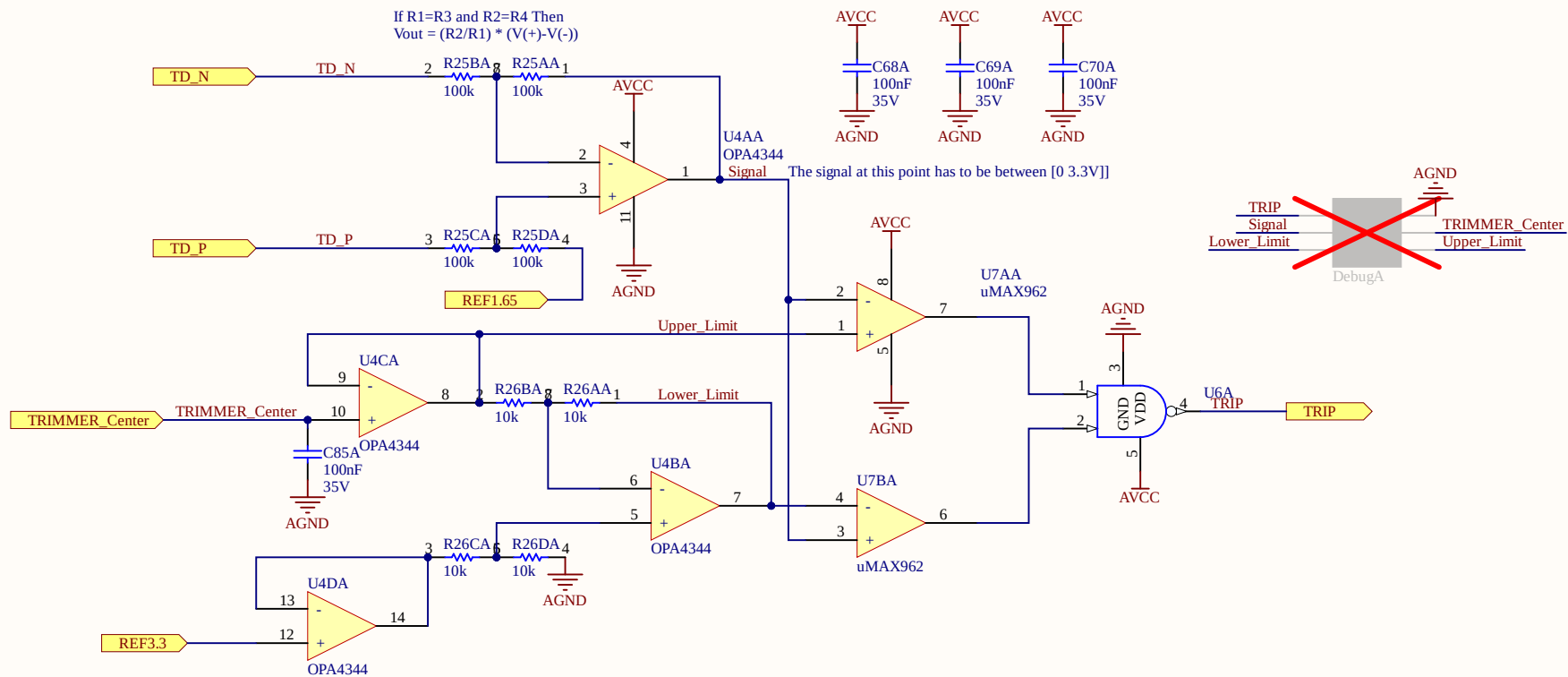
B

C

C

D

D



A

A

B

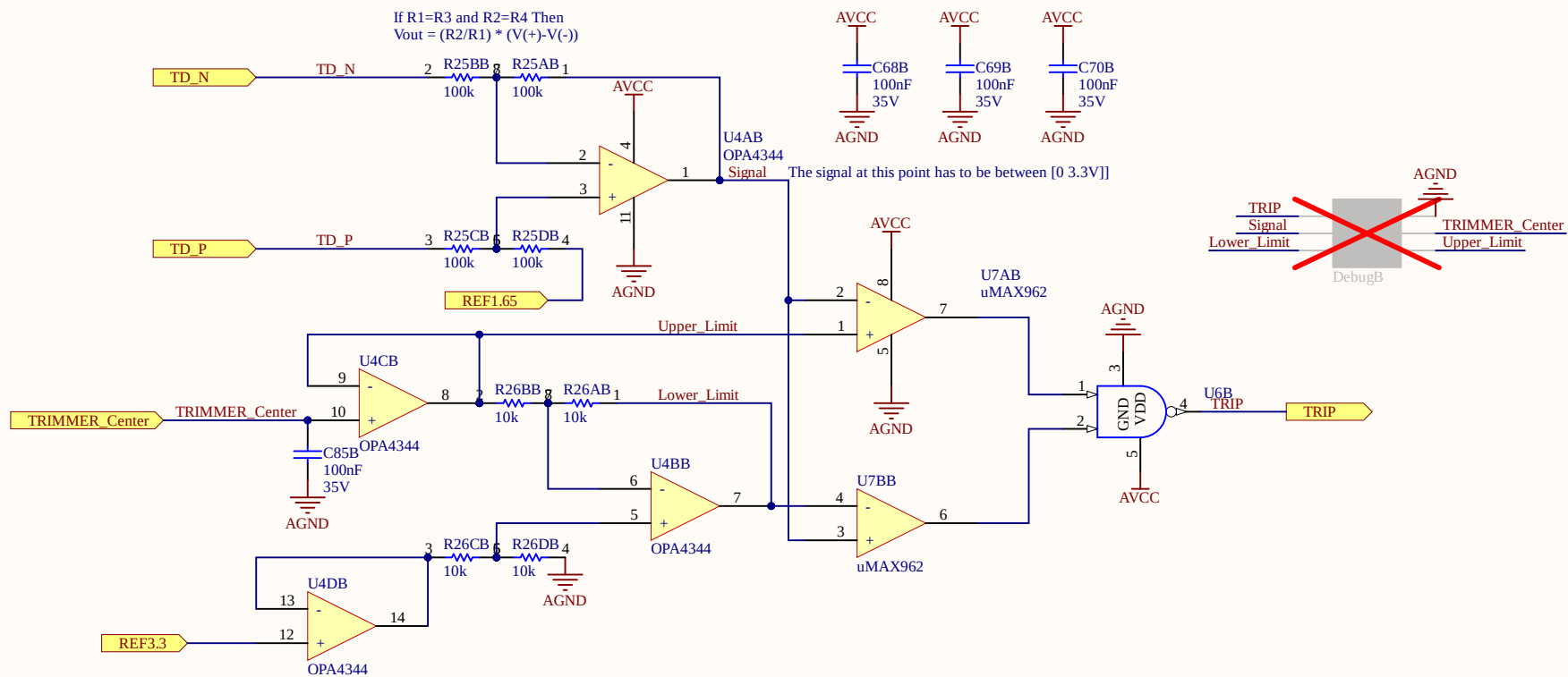
B

C

C

D

D



A

A

B

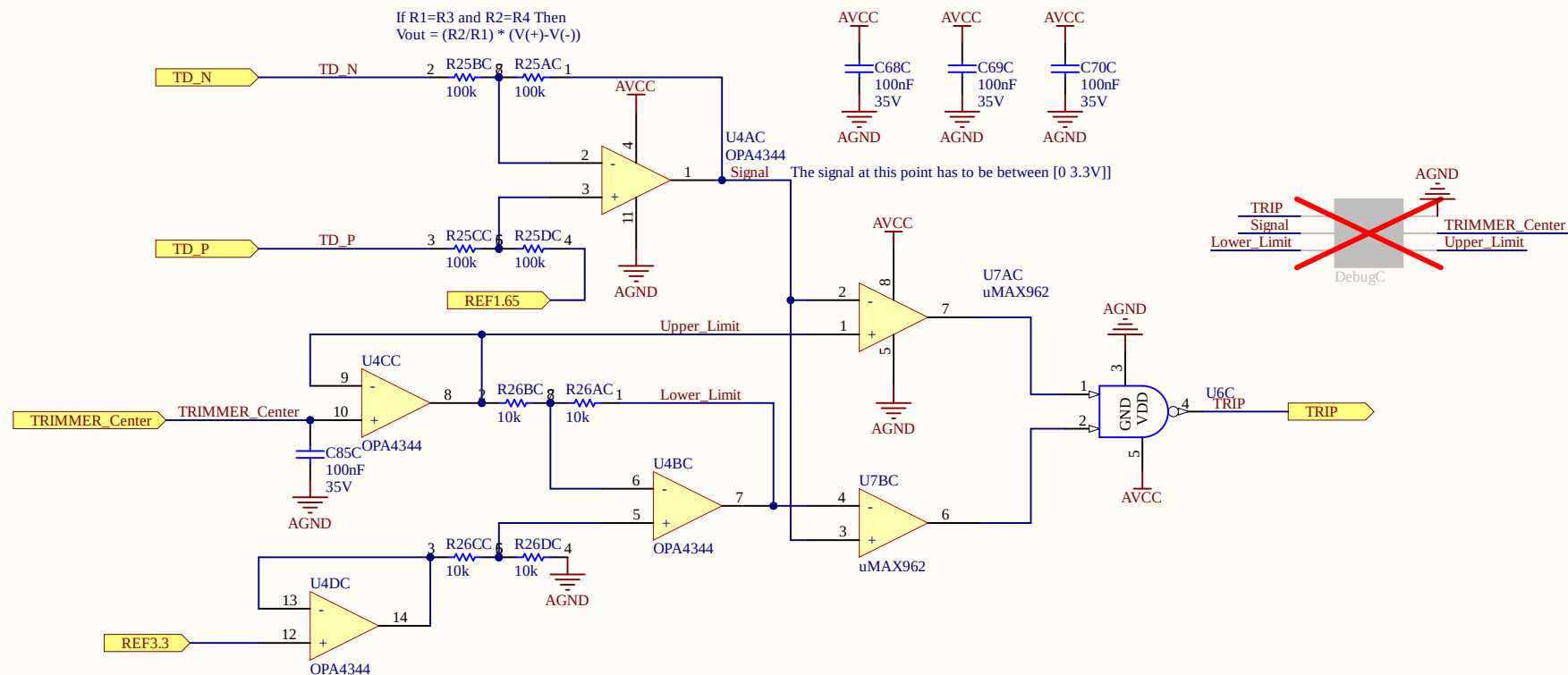
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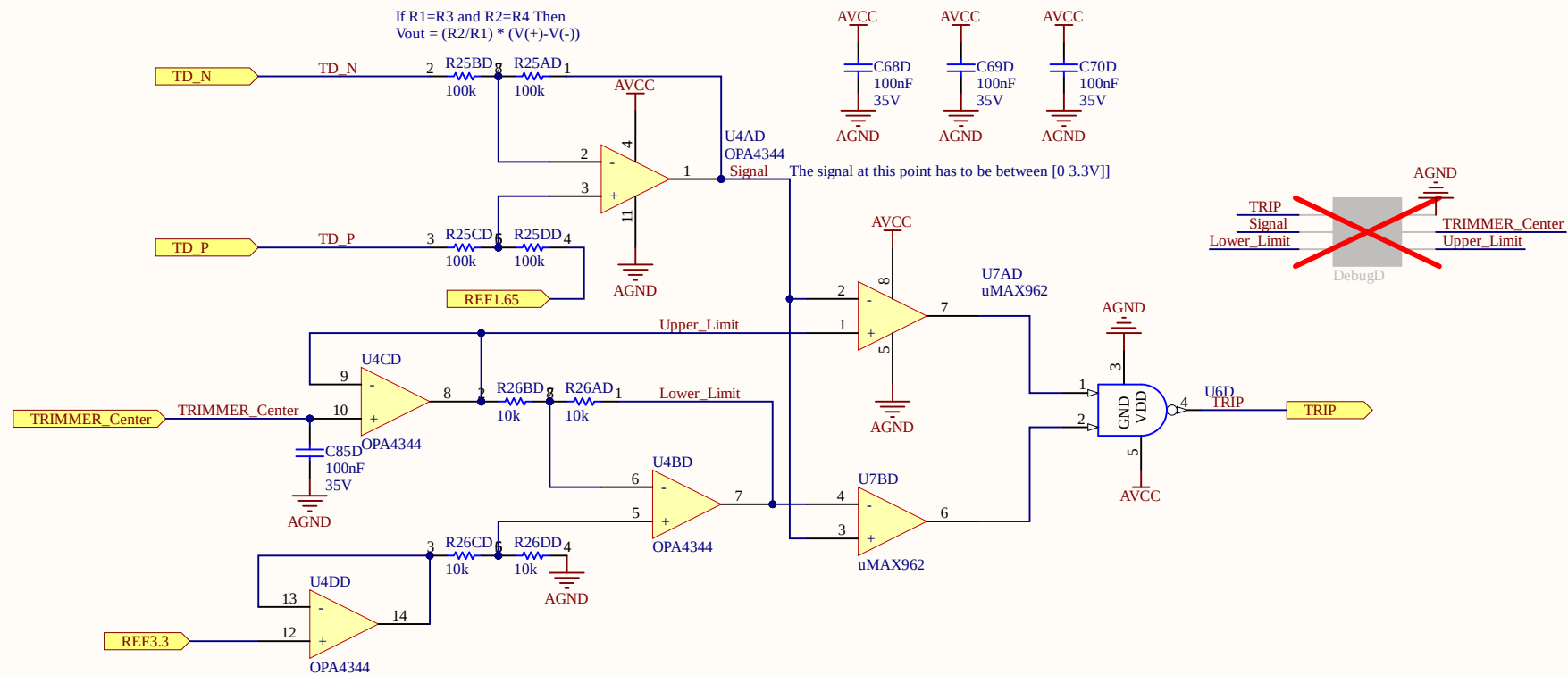
C

C

D

D





A

A

B

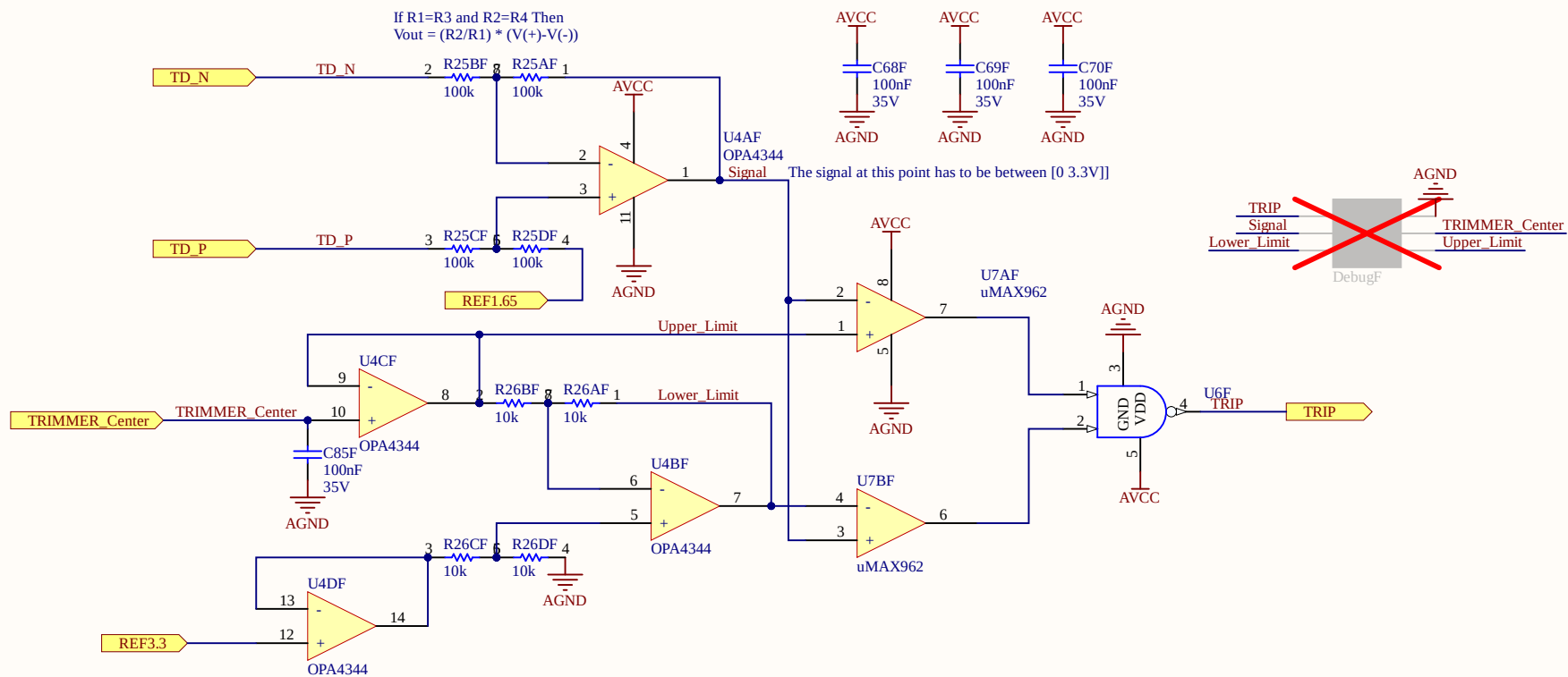
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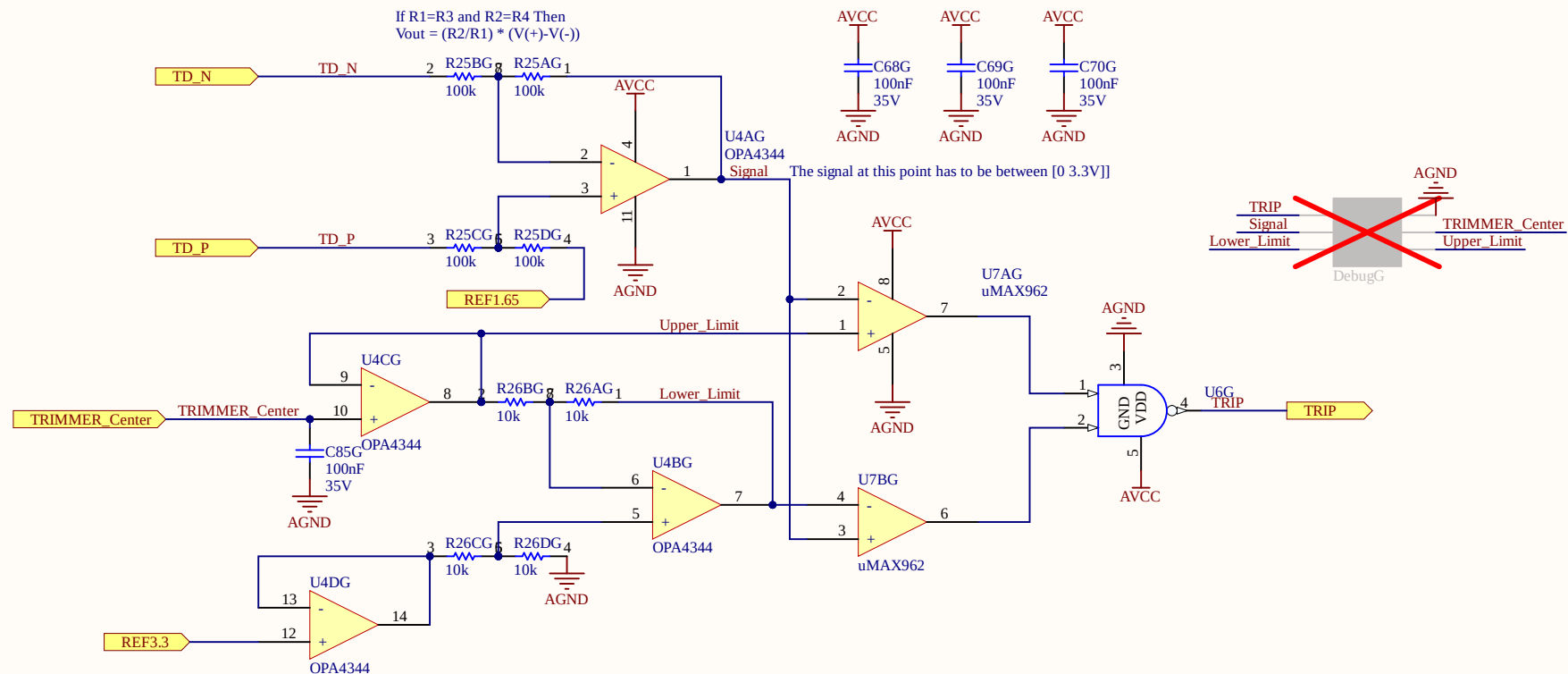
C

C

D

D





A

A

B

B

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C

D

D

